EE426 / 626 Microprocessor Applications

Digital Representation of Data



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Representation	C (gcc)	Assembler (gas)
Character	`'A'`	`.byte 'A'`
String	`"EE426"`	`.asciz "EE426"`
Decimal, unsigned integer	`42U`	`.byte 42`
Decimal, signed integer	`-42`	`.byte -42`
Decimal, float	`3.14f`	`.float 3.14`
Decimal, double	`3.14`	`.double 3.14`
Binary	`0b00101010`	`.byte 0b00101010`
Octal	`052`	`.byte 052`
Hexadecimal	`0x2A`	`.byte 0x2A`
Pointer	`(uint32_t *)0x40020000`	`.word 0x40020000`



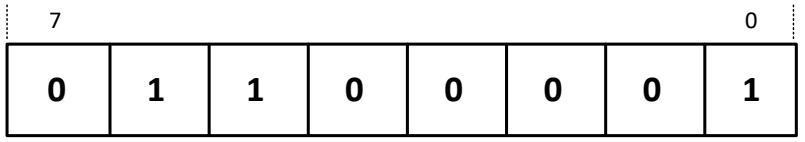
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ASCII Character Mapping

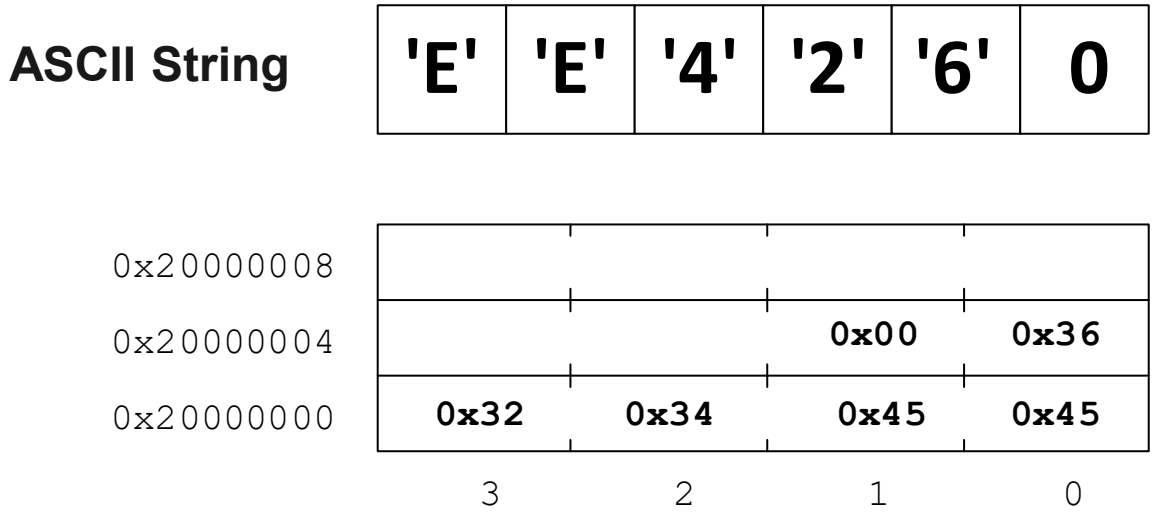
b_4	b_3	b_2	b_1	Row / Column	0 (000)	1 (001)	2 (010)	3 (011)	4 (100)	5 (101)	6 (110)	7 (111)
0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
0	0	1	0	2	STX	DC2	"	2	B	R	b	r
0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
0	1	0	0	4	EOT	DC4	$	4	D	T	d	t
0	1	0	1	5	ENQ	NAK	%	5	E	U	e	u
0	1	1	0	6	ACK	SYN	&	6	F	V	f	v
0	1	1	1	7	BEL	ETB	´	7	G	W	g	w
1	0	0	0	8	BS	CAN	(	8	H	X	h	x
1	0	0	1	9	HT	EM	)	9	I	Y	i	y
1	0	1	0	10	LF	SUB	*	:	J	Z	j	z
1	0	1	1	11	VT	ESC	+	;	K	[	k	{
1	1	0	0	12	FF	FS	,	<	L	\	l	¦
1	1	0	1	13	CR	GS	—	=	M	]	m	}
1	1	1	0	14	SO	RS	.	>	N	^	n	~
1	1	1	1	15	SI	US	/	?	O	_	o	DEL

```
char 'a'


.byte 'a'
.byte 97
.byte 0x61
.byte 0b01100001
```



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```
char str[] = "EE426";          /* array: compiler sizes it automatically -> 6 bytes
                                     (E, E, 4, 2, 6, '\0'), stored in RAM, writable */

const char *str = "EE426";     /* pointer: the literal itself typically lives in
                                     flash (.rodata), str just holds its address */
```

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```
/* --- "array" form: char str[] = "EE426";  (writable copy, lives in RAM) --- */

      .section .data
      .align 2
str:
      .asciz "EE426"

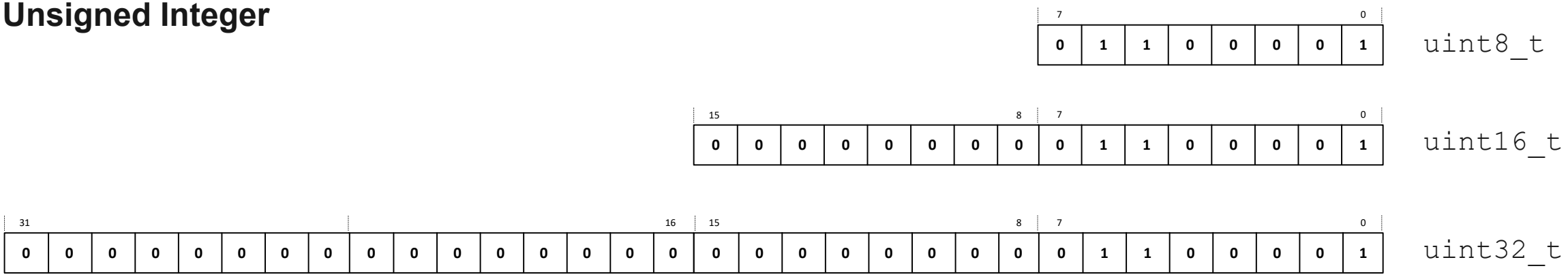

/* --- (literal stays in flash) --- */

      .section .rodata
      .align 2
str:
      .asciz "EE426"
```



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Unsigned Integer

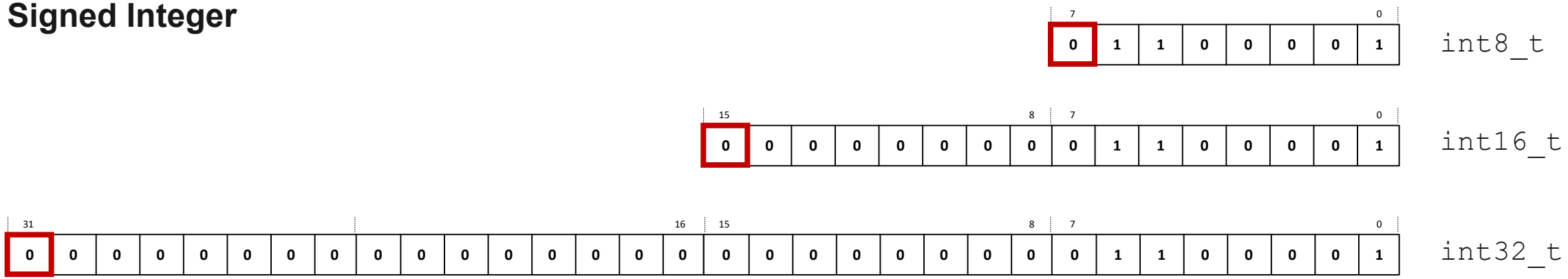


`uint8_t`	0 to 255 (0x00 to 0xFF)
`uint16_t`	0 to 65535 (0x0000 to 0xFFFF)
`uint32_t`	0 to 4,294,967,295 (0x00000000 to 0xFFFFFFFF)



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Signed Integer



`int8_t`	–127 to +127 (0x80 to 0x7F)
`int16_t`	–32768 to +32767 (0x8000 to 0x7FFF)
`int32_t`	–2,147,483,648 to +2,147,483,647 (0x80000000 to 0x7FFFFFFF)

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Simple Binary Coded Decimal BCD (or SBCD)

4-bit BCD codes and pseudo-tetrades

Bit	Weight	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
4	8	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
3	4	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
2	2	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
Name		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
8421 (XS-0)		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

For example – **11:25 am**

Packed BCD

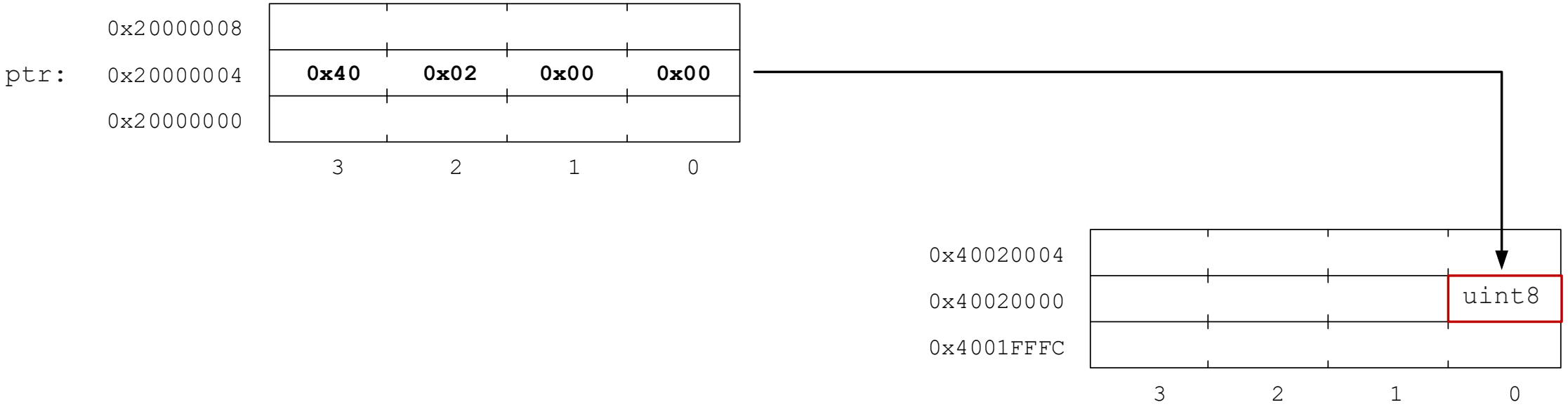
```
0001 0001 0010 0101
```

Unpacked BCD

```
00000001 00000001 00000010 00000101
```

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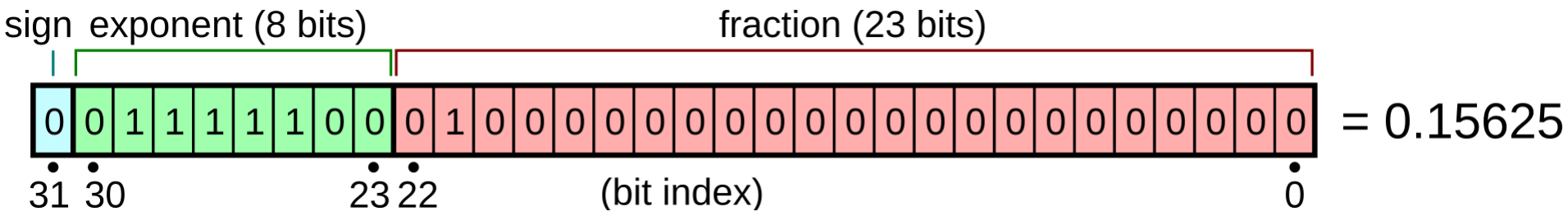
Pointer




```
volatile uint8_t *ptr = (volatile uint8_t *)0x40020000;
```

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Single Precision Floating Point



$$(-1)^{\text{sign}}(1.b_{22}b_{21}\ldots b_0) \times 2^{\text{exp}-127}$$

Minimum = 1.175494×10^{-38}

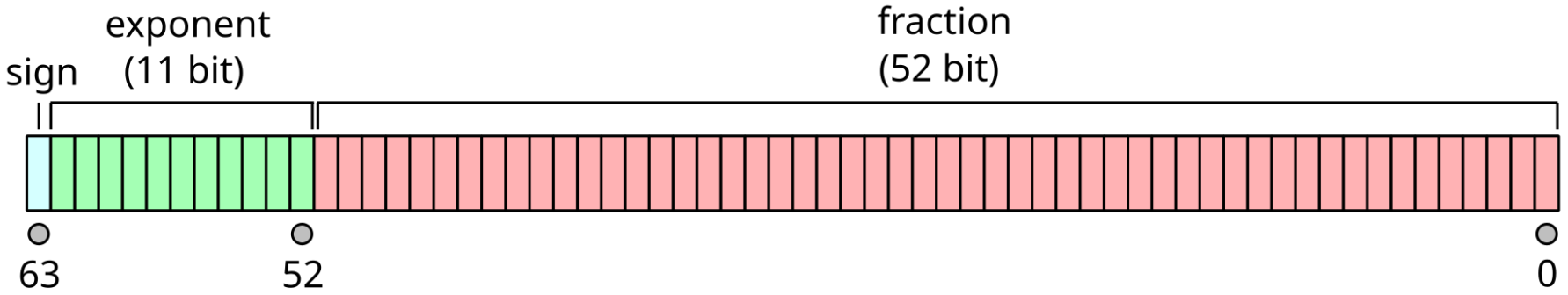
Minimum Subnormal = 1.401298×10^{-45}

Maximum = $3.402823 \times 10^{+38}$

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Double Precision Floating Point



$$(-1)^{\text{sign}}(1.b_{51}b_{50}\ldots b_0) \times 2^{\text{exp}-1023}$$

Minimum $\cong 2.225074 \times 10^{-308}$

Maximum $\cong 1.797693 \times 10^{+308}$

Minimum Subnormal $\cong 4.940656 \times 10^{-324}$

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